# STAT 2020 Discussion Slides

## Week 9: Normal Distribution, Linear Combinations, and Sampling Distribution

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# Index (Outline)


1. Week 9 Goals
2. Standardization and Z-table
3. Percentiles
4. Linear Combinations
5. Linear Expression Probability
6. Sampling Distribution of Xbar

# Agenda

- Normal distribution review and Z-table standardization
- Percentiles: find quantiles, and solve for $\mu$ or $\sigma$
- Mean and variance of linear combinations (assume independence)
- Probability with linear expressions (for example, $P(4X > Y + 1)$)
- Sampling distribution of $\bar{X}$, sums, and related percentiles
- Attendance check

# Standardization Formula

If $X \sim \mathrm{N}(\mu, \sigma^2)$, define

$$Z = \frac{X - \mu}{\sigma} \sim \mathrm{N}(0, 1).$$

Then

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Z-table usually reports $\Phi(z) = P(Z \leq z)$.

# Tail and Interval Forms

$$P(X > x) = 1 - \Phi\left(\frac{x-\mu}{\sigma}\right).$$

$$P(a < X < b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

Common symmetry:

$$\Phi(-z) = 1 - \Phi(z).$$

# Worked Example: Probability Between Two Values

Let $X \sim \mathrm{N}(20, 4^2)$. Find $P(15 < X < 26)$.

$$z_1 = \frac{15-20}{4} = -1.25, \qquad z_2 = \frac{26-20}{4} = 1.5.$$

$$P(15 < X < 26) = \Phi(1.5) - \Phi(-1.25).$$

Using Z-table:

$$\Phi(1.5) \approx 0.9332, \qquad \Phi(-1.25) \approx 0.1056.$$

$$P(15 < X < 26) \approx 0.8276.$$

# Percentile Formula

The $p$-th percentile $x_p$ solves $P(X \leq x_p) = p$.
For $X \sim \mathrm{N}(\mu, \sigma^2)$:

$$x_p = \mu + z_p \sigma,$$

where $z_p$ is the $p$-th percentile of $\mathrm{N}(0, 1)$.
Common value:

$$z_{0.975} \approx 1.96.$$

# Example: Find the 97.5th Percentile

Suppose $X \sim \mathrm{N}(150, 12^2)$. Find $x_{0.975}$.

$$x_{0.975} = 150 + 1.96 \times 12 = 173.52.$$

So the 97.5th percentile is about

$$173.5.$$

# Example: Solve for $\mu$ or $\sigma$

If $x_{0.975} = 180$ and $\sigma = 12$, then

$$180 = \mu + 1.96(12) \quad \Longrightarrow \quad \mu = 156.48.$$

If $\mu$ is known instead, solve

$$\sigma = \frac{x_p - \mu}{z_p}.$$

# Linear Combination Rules

Let $Y = aX_1 + bX_2 + c$.

$$\mathbb{E}(Y) = a\mathbb{E}(X_1) + b\mathbb{E}(X_2) + c.$$

If $X_1 \perp\!\!\!\perp X_2$,

$$\mathbb{V}(Y) = a^2\mathbb{V}(X_1) + b^2\mathbb{V}(X_2).$$

More general form:

$$\mathbb{V}(aX_1 + bX_2) = a^2\mathbb{V}(X_1) + b^2\mathbb{V}(X_2) + 2ab\mathrm{Cov}(X_1, X_2).$$

# Example: $Y = 0.8X_1 + 0.2X_2$

Suppose $X_1, X_2 \overset{\text{iid}}{\sim} \mathrm{N}(20, 4^2)$, and

$$Y = 0.8X_1 + 0.2X_2.$$

$$\mathbb{E}(Y) = 0.8(20) + 0.2(20) = 20.$$

$$\mathbb{V}(Y) = 0.8^2(16) + 0.2^2(16) = 10.88.$$

Since a linear combination of independent normals is normal,

$$Y \sim \mathrm{N}(20, 10.88).$$

# How to Handle $P(4X > Y + 1)$-Type Questions

Rewrite the inequality as one normal variable.
Define

$$W = 4X - Y.$$

Then

$$P(4X > Y + 1) = P(W > 1).$$

If $X, Y$ are independent normals, then $W$ is also normal.

# Example: Compute $P(4X > Y + 1)$

Let $X \sim \mathrm{N}(2, 1^2)$, $Y \sim \mathrm{N}(5, 2^2)$, independent.
Set $W = 4X - Y$. Then

$$\mathbb{E}(W) = 4(2) - 5 = 3, \qquad \mathbb{V}(W) = 16(1^2) + 2^2 = 20.$$

So $W \sim \mathrm{N}(3, 20)$, and

$$P(4X > Y + 1) = P(W > 1) = 1 - \Phi\left(\frac{1-3}{\sqrt{20}}\right) = \Phi(0.447) \approx 0.672.$$

# Theorem for $\bar{X}$

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2)$, then

$$\bar{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Therefore

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0, 1).$$

Also,

$$S_n = \sum_{i=1}^{n} X_i \sim \mathrm{N}(n\mu, n\sigma^2).$$

# Example: Probability for a Sample Mean

Suppose $X_i \overset{\text{iid}}{\sim} \mathrm{N}(20, 4^2)$, $n = 25$. Find $P(\bar{X} > 21)$.

$$\bar{X} \sim \mathrm{N}\left(20, \frac{16}{25}\right), \qquad \mathrm{SE}(\bar{X}) = \frac{4}{5} = 0.8.$$

$$P(\bar{X} > 21) = 1 - \Phi\left(\frac{21 - 20}{0.8}\right) = 1 - \Phi(1.25) \approx 0.1056.$$

# Percentiles of $\bar{X}$ and $S_n$

For $\bar{X} \sim \mathrm{N}(\mu, \sigma^2/n)$:

$$(\bar{X})_p = \mu + z_p \frac{\sigma}{\sqrt{n}}.$$

For $S_n \sim \mathrm{N}(n\mu, n\sigma^2)$:

$$(S_n)_p = n\mu + z_p \sqrt{n}\,\sigma.$$

Same logic: convert to $Z$, use Z-table, then convert back.

# Practice 1: Standardization

If $X \sim \mathrm{N}(60, 8^2)$, find $P(X > 72)$.

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If $X \sim \mathrm{N}(60, 8^2)$, find $P(X > 72)$.

$$z = \frac{72 - 60}{8} = 1.5, \qquad P(X > 72) = 1 - \Phi(1.5) \approx 0.0668.$$

# Practice 2: Percentile

If $X \sim \mathrm{N}(100, 15^2)$, find the 90th percentile.

# Practice 2: Percentile

If $X \sim \mathrm{N}(100, 15^2)$, find the 90th percentile. Use $z_{0.90} \approx 1.2816$:

$$x_{0.90} = 100 + 1.2816(15) = 119.22.$$

# Practice 3: Linear Combination Moments

Let $X_1, X_2 \overset{\text{iid}}{\sim} \mathrm{N}(10, 3^2)$, and $Y = 2X_1 - X_2 + 5$. Find $\mathbb{E}(Y)$ and $\mathbb{V}(Y)$.

# Practice 3: Linear Combination Moments

Let $X_1, X_2 \overset{\text{iid}}{\sim} \mathrm{N}(10, 3^2)$, and $Y = 2X_1 - X_2 + 5$. Find $\mathbb{E}(Y)$ and $\mathbb{V}(Y)$.

$$\mathbb{E}(Y) = 2(10) - 10 + 5 = 15, \qquad \mathbb{V}(Y) = 2^2(9) + (-1)^2(9) = 45.$$

# Practice 4: Linear Expression Probability

If $X \sim \mathrm{N}(1,1)$, $Y \sim \mathrm{N}(2,4)$, independent, compute $P(4X > Y + 1)$.

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If $X \sim \mathrm{N}(1,1)$, $Y \sim \mathrm{N}(2,4)$, independent, compute $P(4X > Y + 1)$.

$$W = 4X - Y \sim \mathrm{N}(2, 20),$$

$$P(4X > Y + 1) = P(W > 1) = 1 - \Phi\left(\frac{1-2}{\sqrt{20}}\right) = \Phi(0.224) \approx 0.589.$$

# Practice 5: Sampling Distribution

If $X_i \overset{\text{iid}}{\sim} \mathrm{N}(50, 9^2)$, $n = 36$, find $P(\bar{X} < 52)$.

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If $X_i \overset{\text{iid}}{\sim} \mathrm{N}(50, 9^2)$, $n = 36$, find $P(\bar{X} < 52)$.

$$\mathrm{SE}(\bar{X}) = \frac{9}{\sqrt{36}} = 1.5, \qquad z = \frac{52 - 50}{1.5} = 1.333.$$

$$P(\bar{X} < 52) = \Phi(1.333) \approx 0.9082.$$

# Class Reminder: Exam 2 Questions

- We can review concepts and methods from homework and quizzes.
- Please do not discuss any actual Exam 2 question in class.
- Students can discuss exam-specific questions during office hours.

# Take-Home Messages

- Standardize first; then use the Z-table.
- Percentiles follow $x_p = \mu + z_p\sigma$ (or its inverse forms).
- For independent variables, linear-combination variance adds by squared coefficients.
- Convert expressions and sample means into one normal variable before computing probability.

# In-Class Attendance Check (Quick)

Suppose $X_1, \ldots, X_{25} \overset{\text{iid}}{\sim} \mathrm{N}(70, 10^2)$.
What is

$$P(\bar{X} < 73)?$$

*Write your answer in $\Phi(\cdot)$ form only (no decimal approximation).*

*Please submit: Full Name; Section Number; Your Answer*

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$$P(\bar{X} < 73)?$$

*Write your answer in $\Phi(\cdot)$ form only (no decimal approximation).*

*Please submit: Full Name; Section Number; Your Answer*

**Answer:**

$$\bar{X} \sim \mathrm{N}\left(70, \frac{100}{25}\right), \quad z = \frac{73 - 70}{10/5} = 1.5,$$

$$P(\bar{X} < 73) = \Phi(1.5).$$